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Programme: B. Tech (DS)/B. Tech Intg. (DS)/MBA Tech (DS)
Year: II / IV/Semester III/Semester VII(Exam Year: 2025-2026)
Subject: Managing Uncertainty
Date: 29 Nov 2025 **Time: 10:00 am to 01:00 pm (03:00 Hrs.)**
Max. Marks: 100

FINAL EXAMINATION (Acad. Year:**2025-2026**)

Instructions:

1. This question paper contains 4 pages.
2. Question No. 1 is compulsory.
3. Out of remaining questions, attempt any 4 questions.
4. In all 5 questions to be attempted.
5. All questions carry equal marks.
6. Answer to each new question to be started on a fresh page.
7. Figure in brackets on the right-hand side indicates full marks
8. Assume Suitable data if necessary.

<table>
<tr><td>Q1</td><td>Q.1</td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-3 ; BL-3</td><td>a.</td><td>The following table gives the frequency distribution of the weekly wages (in '00 Rs.) of 100 workers in a factory.
<table>
<tr><td>Weekly Wages (₹00)</td><td>20–24</td><td>25–29</td><td>30–34</td><td>35–39</td><td>40–44</td><td>45–49</td><td>50–54</td><td>55–59</td><td>60–64</td></tr>
<tr><td>Number of Workers</td><td>4</td><td>5</td><td>12</td><td>23</td><td>31</td><td>10</td><td>8</td><td>5</td><td>2</td></tr>
</table>
Develop the histogram and frequency polygon of the distribution.</td><td>[5]</td></tr>
<tr><td>CO-1,5 ; BL-2</td><td>b.</td><td>Compare Binomial and Poisson Distribution</td><td>[5]</td></tr>
<tr><td>CO-2 ; BL-3</td><td>c.</td><td>A bag contains 7 white and 9 black balls. Two balls are drawn in succession without replacement at random. Identify the probability that one of them is white and the other is black</td><td>[5]</td></tr>
<tr><td>CO-1 ; BL-2</td><td>d.</td><td>Explain the Qualitative and Quantitative types of classification in detail</td><td>[5]</td></tr>
</table>

Q2.a CO-3,4 ; BL-4		Distinguish Karl Pearson's coefficient of skewness from the following data and explain its significance Wages (₹): 70–80 | 80–90 | 90–100 | 100–110 | 110–120 | 120–130 | 130–140 | 140–150 No. of Persons: 12 | 18 | 35 | 42 | 50 | 45 | 20 | 8	[10]
Q2.b CO-2 ; BL-4		Simplify and State Bayes' theorem and provide a detailed proof along with an explanation of its components and applications.	[10]
Q3 CO-3 ; BL-5		In the following data, two class frequencies are missing. Class Interval: 100-110 | 110-120 | 120-130 | 130-140 | 140-150 | 150-160 | 160-170 | 170-180 | 180-190 | 190-200 Frequency: 4 | 7 | 15 | f1 | 40 | f2 | 16 | 10 | 6 | 3 However, it was possible to ascertain that the total number of frequencies was 150 and that the median has been correctly found out as 146·25. You are required to Evaluate with the help of information given : (i) The two missing frequencies. (ii) Having found the missing frequencies, calculate arithmetic mean and standard deviation. (iii) Find the value of mode.	[20]
Q4.a CO-2,6 ; BL-5		The joint probability function of X and Y is given by $p(x,y)=k(x^2+y)$ for x=1,2 and y=1,2,3. Determine: (i) The value of k. (ii) The marginal distributions of X and Y. (iii) Are X and Y independent? (iv) The conditional distribution of X for Y=2.	[10]
Q4.b CO-2 ; BL-3		Ten unbiased coins are tossed simultaneously. Identify the probability of obtaining; (i) Exactly 6 heads (ii) At least 8 heads (iii) No head (iv) At least one head (v) Not more than three heads (vi) At least 4 heads.	[10]
Q5.a CO-2,5 ; BL-3		The time (in minutes), a customer spends waiting for a service agent at a call center follows an Exponential Distribution with a mean waiting time of 5 minutes. (a) Determine the probability density function (PDF) and the cumulative distribution function (CDF) for the waiting time, X. (b) Find the probability that a customer waits for more than 8 minutes. (c) Find the probability that a customer waits between 3 and 7 minutes.	[10]

<table>
<tr><td>Q5.b
CO-2 ;
BL-4</td><td></td><td>There are three identical-looking boxes: B1, B2, and B3. The composition of the balls in each box is as follows:
Box B1: 5 Red balls, 5 Green balls
Box B2: 2 Red balls, 8 Green balls
Box B3: 7 Red balls, 3 Green balls

A box is chosen at random, and then a ball is drawn from that box. The drawn ball is observed to be Red. Analyze is the probability that the ball came from Box B3?</td><td>[10]</td></tr>
<tr><td>Q6.a
CO-3,6 ;
BL-2</td><td></td><td>Explain Type I and Type II error. Explain Skewness and Kurtosis in detail</td><td>[10]</td></tr>
<tr><td>Q6.b
CO-3 ;
BL-3</td><td></td><td>The number of critical errors detected per hour on a new e-commerce website was observed over a 200-hour testing period.

<table><tr><td>Number of Errors per Hour (X)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Observed Frequency (f)</td><td>120</td><td>55</td><td>18</td><td>5</td><td>2</td></tr></table>
Identify the expected frequencies using the Poisson distribution. Also, compute the mean and variance of the fitted (theoretical) distribution.</td><td>[10]</td></tr>
<tr><td>Q7.a
CO-2;
BL-5</td><td></td><td>A continuous random variable X has the following probability density function
$$f(x) = \frac{a}{x^5}, 2 \leq X \leq 10$$
Determine the constant a, distribution function of X and find the probability of the event $4 \leq X \leq 7$</td><td>[10]</td></tr>
<tr><td>Q7.b
CO-5,6 ;
BL-6</td><td></td><td>A manufacturer claims that the mean lifespan of their new LED bulb is 8,000 hours. A consumer testing agency randomly selected 12 bulbs and recorded their actual lifespans (in hundreds of hours):

78,85,75,82,79,88,80,81,74,86,83,76

Test whether these results support the manufacturer's claim? Use a significance level of α=0.01.</td><td>[10]</td></tr>
</table>

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										